

osu!Stocks - Hosting & Vendor Recommendation

Table of Contents

Hosting & Vendor Recommendation	1
TL;DR — recommendation	1
What we're actually hosting.....	2
Option tiers	2
Recommended vendor shortlist (by category)	4
12-month cost projection (recommended Option A).....	4
When to upgrade	5
Notes for the sponsor.....	5

Hosting & Vendor Recommendation

Audience: sponsor / budget owner. Purpose: pick where osu!Stocks runs in production and what it costs per month. Pricing checked **June 2026** (sources at the bottom). Treat as indicative — confirm at signup. Hetzner prices are **ex-VAT** in EUR; USD figures are approximate conversions.

TL;DR — recommendation

Run the whole stack on **one small VPS (Hetzner)** using the Docker Compose setup we already have. Nothing needs to be re-architected.

Recommended host	Hetzner Cloud CX32 (4 vCPU / 8 GB / 80 GB)
Domain	.com via Cloudflare / Porkbun / Spaceship
SSL	Free (Let's Encrypt)
Off-host backups	Cloudflare R2 or Backblaze B2 (free tier covers us)
Uptime monitoring	UptimeRobot (free)

≈ **Total** ~\$13–15 / month (≈ \$155–180 / year)

This is the cheapest credible production setup. We can move to managed databases or a platform-as-a-service later if reliability needs grow — see the tiers below.

What we're actually hosting

The app is five Docker containers defined in `docker-compose.yml`:

Container	Role	Resource profile
api	ASP.NET Core API (public)	light–moderate
worker	Background jobs (sync, market engine)	light, periodic
postgres	PostgreSQL 16 — the only durable data	needs RAM + reliable disk
redis	Cache / rate-limit / OAuth state	light
nginx	Reverse proxy + TLS termination	light

So this is a **single modest server's** worth of work at launch. **8 GB RAM** is the comfortable size (4 GB works but is tight running .NET + Postgres + Redis together). The key risk to protect against is **Postgres data loss**, which the backup plan below handles.

Option tiers

★ Option A — Single VPS, self-hosted (recommended for launch)

Everything in one VPS via `docker compose up`. Cheapest, matches our current setup exactly.

Item	Vendor / plan	Monthly
VPS (4 vCPU / 8 GB / 80 GB)	Hetzner CX32	€8.99 ≈ \$9.80
Domain (.com, amortised)	Cloudflare/Porkbun (~\$11/yr)	~\$0.90
TLS certificate	Let's Encrypt	\$0
Off-host DB backups	Cloudflare R2 / Backblaze B2 free tier	~\$0
Uptime monitoring	UptimeRobot free	\$0

Item	Vendor / plan	Monthly
Total		≈ \$11–13 / month

-  Lowest cost, full control, no vendor lock-in, identical to dev setup.
-  We run OS patching + the DB ourselves. Mitigated by automated backups (see OPERATIONS_RUNBOOK.md §6) and Hetzner's optional auto-snapshots (+20% of server price ≈ \$2/mo).
- **Cheaper variant:** Hetzner **CX22** (2 vCPU / 4 GB) at €4.49 ≈ **\$4.90/mo** → total **~\$6–8/mo**. Fine to start; size up if memory gets tight.

Option B — VPS + managed PostgreSQL (more resilient)

Keep app on a VPS, move the database to a managed service (automated backups, failover, patching handled for us).

Item	Vendor / plan	Monthly
VPS (app + worker + redis + nginx)	Hetzner CX22	~\$4.90
Managed PostgreSQL (4 GB)	DigitalOcean Managed Postgres	\$60.90
Domain + TLS + monitoring	as above	~\$1
Total		≈ \$67 / month

-  Database is someone else's problem (backups, HA, upgrades).
-  ~5× the cost of Option A. Worth it only once data/users justify it.

Option C — Platform-as-a-Service (least ops, no server)

Push containers; the platform runs them. Easiest to operate, priciest per unit.

Item	Render	Railway
API (web service)	\$7–25	usage-based
Worker (background)	\$7+	usage-based
PostgreSQL	from \$6	~\$92
Redis	add-on	add-on
Rough total	~\$25–50/mo	\$20–100/mo (scales with usage)

-  No Linux/Docker host to maintain; fast to ship; easy scaling.
-  More expensive at steady load; some lock-in; need to split our compose into per-service definitions.

Recommended vendor shortlist (by category)

VPS — pick one: | Vendor | Entry plan (≈8 GB) | Monthly | Notes | |---|---|---|---| |

Hetzner ★ | CX32 (4 vCPU/8 GB) | €8.99 ≈ \$9.80 | Best price/performance; EU + US regions | | DigitalOcean | 8 GB / 2 vCPU droplet | ~\$48 | Polished UX, more expensive | | Vultr / Linode | 8 GB | ~\$40–48 | Comparable to DO |

Domain registrar — .com ~\$9–12/yr, all include free WHOIS privacy: | Vendor | New | Renewal | |---|---|---| | **Cloudflare** ★ | at-cost | ~\$10.44/yr (transfer-in only) | | Porkbun | ~\$11.08 | ~\$11.08 | | Spaceship | ~\$9.08 | ~\$10.18 |

SSL: Let's Encrypt — **free**, automated via certbot or Caddy. No paid cert needed.

Off-host backups (critical): Docker volumes don't survive host loss. Push the daily pg-backup.sh dump to object storage: | Vendor | Free tier | Then | |---|---|---| |

Cloudflare R2 ★ | 10 GB free, no egress fees | \$0.015/GB-mo | | Backblaze B2 | 10 GB free | \$6/TB-mo |

Our DB dump is megabytes → effectively **\$0**.

Uptime monitoring: UptimeRobot (free, 50 monitors) on <https://api.<domain>/health>. Optional: Better Stack free tier for logs + status page.

12-month cost projection (recommended Option A)

Line item	Monthly	Year 1
Hetzner CX32 VPS	~\$9.80	~\$118
Domain (.com)	~\$0.90	~\$11
TLS (Let's Encrypt)	\$0	\$0
Off-host backups (R2/B2 free tier)	~\$0	~\$0
Uptime monitoring (UptimeRobot free)	\$0	\$0
(optional) Hetzner auto-snapshots	~\$2	~\$24
Total	~\$11–13	~\$130–155

Starting on the **CX22 (4 GB)** instead drops this to **~\$6–8/month (~\$75–95/year)**.

When to upgrade

- **Memory pressure / slow queries** → bump the VPS one size (Hetzner resize is a few minutes of downtime).
- **Can't afford DB downtime / want hands-off DB** → move to **Option B** (managed Postgres).
- **Don't want to run a server at all** → move to **Option C** (Render/Railway).
- **Traffic growth** → put Cloudflare (free) in front for CDN/DDoS, then scale the VPS or split services.

We can launch on Option A and revisit only when a real constraint appears — no need to pre-pay for scale.

Notes for the sponsor

- These are **infrastructure** costs only. There are no per-user fees; osu! OAuth and Let's Encrypt are free.
 - One **annual domain renewal** (~\$11) is the only yearly lump sum; everything else is monthly and cancellable.
 - Total exposure to start is roughly **a coffee or two per month**. The biggest operational risk (database loss) is covered by free off-host backups.
 - Recommendation: **start on Hetzner CX22 or CX32 (Option A)**; it mirrors our current Docker setup exactly, so deployment is `docker compose up per OPERATIONS_RUNBOOK.md`.
-

Sources (verify current pricing at signup)

- [Hetzner Cloud pricing](#) · [Hetzner CX22 pricing 2026](#) · [Hetzner price adjustment \(Apr 2026\)](#)
- [DigitalOcean Droplet pricing](#) · [DigitalOcean Managed Databases pricing](#)
- [Render pricing](#) · [Render vs Railway 2026](#)
- [Cheapest domain registrars 2026](#)